

E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS

25J-26-103

Project Proposal Report

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August 2025

DECLARATION

DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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ABSTRACT

AI-driven tools and Large Language Models (LLMs) transformed the nature of education into a personal and accessible experience that cuts across various user groups. Nevertheless, there have been complaints that hearing-impaired students cannot enjoy their studies in most conventional e-learning because the technology does not fully facilitate sign language, emotions, and multisensory communication. It requires continued improvements to result in inclusivity, flexibility and in-time responsiveness regarding the areas of specialization in teaching and learning.

The offered system goes further in e-learning exercises as it designs a comprehensive interface that will fit in hearing-impaired students. With a combination of sign language video generation, emotion tracking, multisensory immersion and lip-reading assistance, aggregated into a coherent system, it is expected to increase the accessibility and interest in taking care of learning queries and tasks. Such a form of engagement not only adds more depth and personalization to learning experiences but also circumvents short-comings inherent to more traditional platforms when scaled to hearing-impaired learning.

Finally, as shown in this study, this can close the gap between universal e-learning solutions and the necessity of culturally acceptable and domain-specific tools in Sri Lanka, leading to more inclusive and accessible uses not only within the education sector.

ACKNOWLEDGMENT

This is my moment to thank all the people who have supported the preparation of the project proposal report of this Inclusive E-Learning Platform to the hearing-impaired students.

First, I would like to appreciate our supervisor, Dr. Lakmini Abeywardhana, who would guide us with the research problem. Her experience and direction in the area of Natural Language Processing (NLP) and Machine learning has played a critical role in assisting us to develop our proposed solution. Then also, I would like to extend my gratitude to our co-supervisor Ms. Sanjeevi Chandrasiri who has been very supportive. Her input, in terms of perfecting and examining our ingredients and running us through lectures on how to be more effective in our research, has been a priceless asset to our growth.

Next, I would wish to express my gratitude to my colleagues and acknowledge their efforts and work diligence they have contributed to the realization of this project since the very beginning. I also personally experienced the level of interest to the research issue and the enthusiasm to come up with ideas and cooperate to find the optimal solution. Their loyalty and passion has been inviting.

Finally, I would also take this opportunity to express my sincere gratitude to the faculty of computing and the comprehensive design and analysis project (CDAP) at the Sri Lanka institute of information technology (SLIIT). I sincerely appreciate the fact that they have allowed me to venture and offer my services in this identity study. The reason is because of their consistent help and resources given to us to organize the project proposal.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DNN	Deep Neural Network
GCN	Graph Convolutional Network
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
RNN	Recurrent Neural Network
3D-CNN	Three-Dimensional Convolutional Neural Network
API	Application Programming Interface
UI	User Interface
UX	User Experience
WebRTC	Web Real-Time Communication
F1 Score	Harmonic mean of Precision and Recall

1. INTRODUCTION

1.1. Background of the Research

Education plays an essential role as a universal right and a prerequisite to personal and societal growth, but yet to the hearing impaired, especially those in a developing country such as Sri Lanka, major challenges still remain within the acquisition of an inclusive and equitable right of education. Estimates confirm that there are around 300,000 deaf or hard of hearing people in Sri Lanka (DHH) in the Sinhala language speaking population, and this is a huge percentage of the population that is dependent on visual communication means when dealing with everyday activities and even schooling. A survey conducted in a district in Gampaha established a prevalence rate of hearing impairment in school children of 5.5 percent with most of them having a mild to severe level that affects their school performance. In the whole world, the world health organization notes that there are currently more than 430 million individuals living with disabling hearing loss and the number is expected to increase to 700 million by the year 2050 confirming the imminence of controlled interventions in areas with poor resources to limit the influence of hearing loss.

In Sri Lanka, the systemic problems of training teachers, lack of specialized resources and overuse of oral methods at the expense of visual-linguistic approach of deaf learners has always been plaguing the education system of DHH children. Regional differences in the use of sign language facilities have occurred in 25 deaf schools that are found within the country, yet standardization of the Sri Lankan Sign Language (SLSL) over the past decades has been to streamline communication. An independent visual language with colonial-era origins and local adaptations, SLSL has an estimated 2020 population of 107,000, and uses its own grammatical forms, hand shapes, and non-manuals including expressions; this means that it can convey meaning not only through gesture, but with what is called additional support. Nevertheless, educational inequality is worsened by issues regarding the absence of standardized signs communicating the idea of complex concepts, acoustic obstacles in the regular classrooms, and the availability of older assistive devices, which results in the low literacy rates of DHH students, and the low rate of continuing education among them. These problems were further escalated by

the COVID-19 pandemic due to the relative neglect of DHH needs on the remote learning platforms, which led to isolation and uneven learning.

The current hearing impaired students e-learning solutions in Sri Lanka are promising in spite of the limited incidence and capabilities. Other apps such as Hastha offer both game-based learning and simple gesture translation between SLSL and American Sign Language and are one-way content-delivery platforms. In a similar manner, EasyTalk uses machine learning to ensure that gestures are interpreted into text or voice and the other way around to allow basic conversion of verbal to sign. Other initiatives like Visual Kids and Esign Guru, have modules interactively used at primary school levels in SLSL, recognizing letters and numbers using visual aids. Regardless of these developments, the majority of the tools focus on passive reception of the content, e.g. by watching pre-recorded signs or having read basic translations, but fail to facilitate real-time bidirectional communication and cultural relevancy to the linguistic concepts of SLSL. It is also lacking features that allow active engagement like ability to correct hand gesture error, integration of facial expression or adaptive avatar focused synthesis that would help the students to engage in dynamic dialog in the classrooms or give them personal feedback. The gap has the overall effect of reinforcing a "free but not fair" educational environment, in which DHH students are usually sidelined when it comes to collaborative learning processes, which has a cascade effect in creating lack of confidence, standardization problems when treating the SLSL, and a low integration into everyday society.

This is an opportunity to fill these gaps, and the fast advancement of artificial intelligence (AI), natural language processing (NLP), and deep learning technologies offer a chance to become closer to closing these gaps. Advances in generative adversarial networks (GANs), transformers, and convolutional neural networks (CNNs) have allowed to perform advanced gesture recognition and synthesis of other sign languages, but the methods adapted to SLSL are still underdeveloped. With the use of these tools, an e-learning platform could change accessibility as adaptation of content to an interactive cultural aware education where actively-participating and skill-affirmed students with hearing difficulties in Sri Lanka exist. This study responds to these

imperative needs by introducing an AI-based system that, besides transcribing lecture materials into SLSL through the involvement of animated avatars, will also encompass real-time communication and coaching on gestures, which will lead to opening the learning process toward inclusive, two-way communication.

1.2. Literature Survey

The construction of e-learning environments to help hearing-impaired students has received serious consideration over the recent past especially the technical progress in the artificial intelligence (AI), computer vision, and natural language processing (NLP). The survey examines available literature pertaining to sign language recognition, generation, avatar based systems, gesture feedback mechanisms, and avatars that could be customized, especially when used within an educational setting. Special consideration is given to the systems of Sri Lankan Sign Language (SLSL), with strengths, as well as gaps, being identified before being addressed by the proposed AI-driven sign language video generation and customization platform.

1.3. Research Gap

The research gap is to investigate the need to develop an advanced e-learning platform that integrates real-time bi-directional communication, AI-powered gesture error correction, and customizable avatar-based synthesis specifically for Sri Lankan Sign Language (SLSL) in educational settings for hearing-impaired students, with a focus on avatar styles (teacher, realistic, cartoon) to enhance engagement. Existing research highlights limitations in interactivity, feedback mechanisms, and cultural adaptation, but none incorporate customizable avatar styles tailored for SLSL learners. This gap is illustrated through the following studies.

Research A: This study introduces a wearable sensor-based system for interpreting SLSL, focusing on static sign recognition for hearing-impaired children in educational contexts [1]. While it achieves moderate accuracy in recognizing isolated signs, it lacks real-time two-way communication, integration of non-manual markers like facial expressions, and customizable avatars to enhance learner engagement, limiting its applicability to dynamic e-learning environments that require active participation and culturally relevant visual aids.

Research B: This research presents EasyTalk, a machine learning-based approach to translate SLSL gestures into Sinhala text or speech, aimed at facilitating basic communication for hearing-impaired individuals [2]. Although it demonstrates effective one-way translation, the system does not support synchronous bi-directional interaction, gesture error correction, or avatar-based synthesis with styles like teacher, realistic, or cartoon, failing to address the need for engaging, standardized SLSL learning in classroom settings.

Research C: This work describes Hastha, a real-time SLSL translation system that employs gamebased learning and basic gesture translation to American Sign Language, enhancing student-system interaction in e-learning [6]. Despite its potential for engaging content delivery, it lacks features for real-time synchronous communication, SLSL-specific non-manual markers, and customizable avatars designed to boost engagement, restricting its ability to foster collaborative, culturally attuned learning experiences for hearing-impaired students in Sri Lanka.

	Research A	Research B	Research C	Preposed
Comprehensive SLSL-specific system with real-time, Bi-directional sign language video generation	✗	✗	✗	✓
Avatar styles (teacher, realistic, cartoon) tailored to enhance engagement for SLSL learners in educational settings.	✗	✗	✗	✓
Detects and corrects gesture inaccuracies (handshape, motion) with interactive visual feedback, limiting learning precision and confidence.	✗	✗	✗	✓

Table 1: Research gap illustration

1.4. Research Problem

The research problem centers on the critical absence of a robust, accessible, and interactive elearning platform specifically designed to meet the educational needs of hearing-impaired students in Sri Lanka, with a focus on Sri Lankan Sign Language (SLSL). Existing systems, such as basic translation tools like EasyTalk or limited avatar-based platforms like SAIL, predominantly cater to American Sign Language (ASL) or offer unidirectional communication, lacking the capability for real-time, bi-directional interaction. This deficiency is compounded by the absence of advanced features such as AI-driven sign language video generation that can dynamically convert lecture content (text or audio) into SLSL videos using 3D avatars, which are essential for an immersive learning experience. Furthermore, these systems do not support customizable avatar styles—such as teacher-like, realistic, or cartoon representations—that could enhance learner engagement and personalization, a feature particularly relevant in diverse educational contexts.

An additional challenge is the lack of sophisticated gesture correction mechanisms. Current tools, such as ASL Champ!, provide rudimentary feedback on sign accuracy but fail to address specific inaccuracies in SLSL gestures, including handshape, motion trajectory, and facial expressions—non-manual markers critical to SLSL. This gap prevents learners from receiving targeted, interactive feedback through visual overlays, slow-motion demonstrations, or highlighted gesture comparisons, which are vital for improving accuracy and building confidence. The absence of a real-time, device-optimized web application further exacerbates accessibility issues, limiting the platform's usability across standard devices in classroom or tutoring settings. Consequently, the current technological landscape leaves hearing-impaired SLSL users without a comprehensive, end-to-end e-learning solution that integrates these innovative features, resulting in suboptimal educational outcomes and reduced participation in mainstream learning environments. This research seeks to address these multifaceted shortcomings to create a culturally relevant, inclusive, and effective learning platform.

2. OBJECTIVES

2.1. Main Objective

To develop an AI-powered e-learning platform tailored for hearing-impaired students in Sri Lanka, enabling real-time bi-directional communication in Sri Lankan Sign Language (SLSL) through natural language processing (NLP), deep learning-based gesture synthesis, and CNN-based gesture recognition, incorporating gesture error correction and 3D avatar's styles to foster active participation, skill improvement, and standardization in educational settings.

2.2. Sub Objectives

1. Develop an AI-Driven Sign Language Video Generation System

- It is proposed to build a powerful natural language processing (NLP) pipeline to transform lecturer materials, given in the form of written text or audio recordings, into top-quality videos of the Sri Lankan Sign Language (SLSL).
- To facilitate fluid inclusion into live teacher-led classroom and tutoring activities, optimize the pipeline to address low-latency requirements, and to enable Sinhala-to-sign and sign-to-Sinhala translation in a scalable manner using middleware. This guarantees culturally pertinent and reachable learning experience to the situation within Sri Lanka.

2. Implement Avatar Styles

- Develop a new functionality that allows one to select the style and easily alternate between the various avatar styles; the teacher-like, realistic and cartoon ones, to foster personalization and meet the needs of different learners. The purpose of this customization is to engage and motivate the hearing-impaired students because their avatars will be made in their preferred style or according to the cultural expectations. Consider the effects of personalized avatars in terms of retention and

satisfaction by testing with users, with the accent on the modeling of these styles consistency with the learning experience of those who use learning tools in SLSL with hearing impairment in an isolated and collaborative study.

3. Establish Gesture Accuracy Detection and Correction Mechanism

- Combine with the latest-state computer vision and artificial intelligence tools to analyze and identify any inaccuracies in the gestures that students use to create signs, including errors in their handshape, direction of motion, facial expression facial expressions- three important elements of SLSL. Use deep learning models to give the exact detection of these discrepancies in real-time.
- Design interactive feedback mechanism which provides feedback by sensory modality i.e. visual overlays of correct and incorrect gestures, slow-motion demonstrations to provide detailed visualization of complex motions .This twopronged strategy will help increase the specificity of learning and increase confidence in the students.

3. METHODOLOGY

This section outlines the methodologies that will be followed throughout the research and development phase to achieve the objectives of the e-learning platform for hearing-impaired students using Sri Lankan Sign Language (SLSL). The aim is to develop a comprehensive web application tailored to the educational needs of hearing-impaired students, leveraging AI-driven sign language video generation, customizable avatar styles, and gesture accuracy feedback. Given the diverse access needs within educational environments in Sri Lanka, where users may utilize desktops, tablets, or smartphones, the project will focus on creating a responsive webbased solution. The development process will be iterative, involving continuous refinement and enhancement of the system to meet the research goals effectively.

3.1. System Architecture Diagram

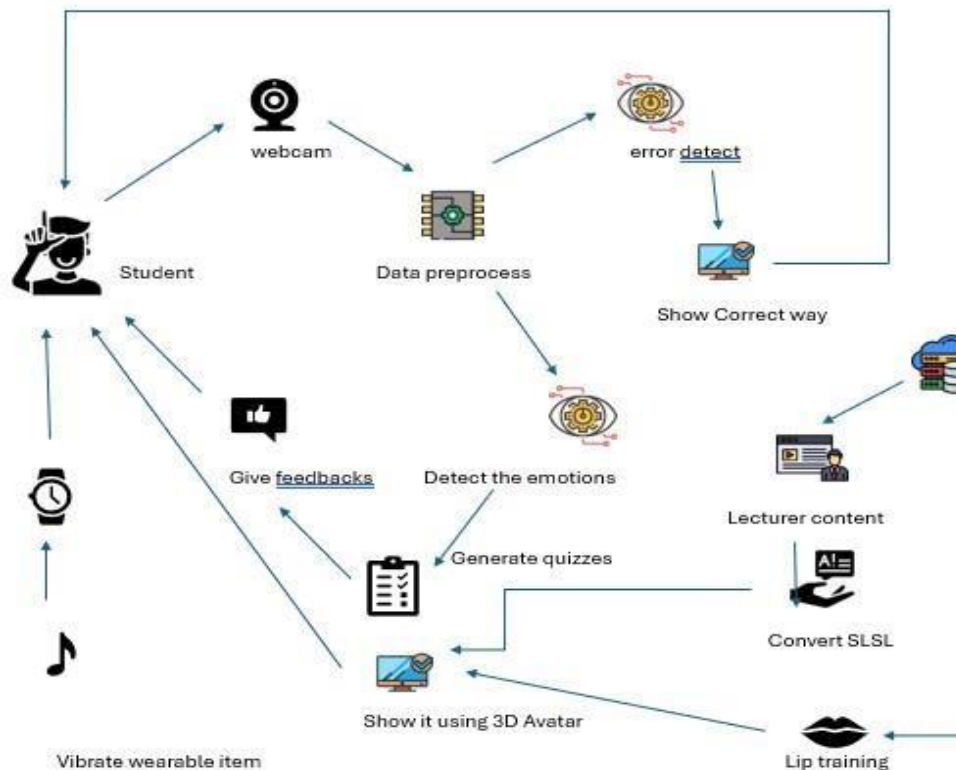


Figure 1: System Architecture Diagram

The process begins with the User Interface (UI) Module, a web-based dashboard tailored for students and educators. This module captures diverse inputs—webcam data (gestures, emotions, lip movements) and lecturer content (text/audio)—and allows customization of avatar styles (teacher, realistic, cartoon). It serves as the entry point, delivering real-time outputs such as SLSL videos, emotion feedback, haptic cues, and progress reports, fostering an interactive and adaptive learning experience.

Inputs are processed by the Input Processing Module, which employs Natural Language Processing (NLP) for text/audio and computer vision for gesture and emotion analysis. This module segments data into culturally relevant formats, feeding structured outputs (phonemes, gesture features, emotion metrics) to subsequent AI-driven components.

The AI-Driven Sign Language Generation Module utilizes Generative Adversarial Networks (GANs), Transformers, and Convolutional Neural Networks (CNNs) to convert processed content into SLSL video sequences. It incorporates facial expressions and non-manual markers, enabling bi-directional communication, and links to the Avatar Customization and 3D Animation Module. This module, powered by Blender and Three.js, animates 3D avatars with customizable styles, ensuring natural and culturally resonant interactions.

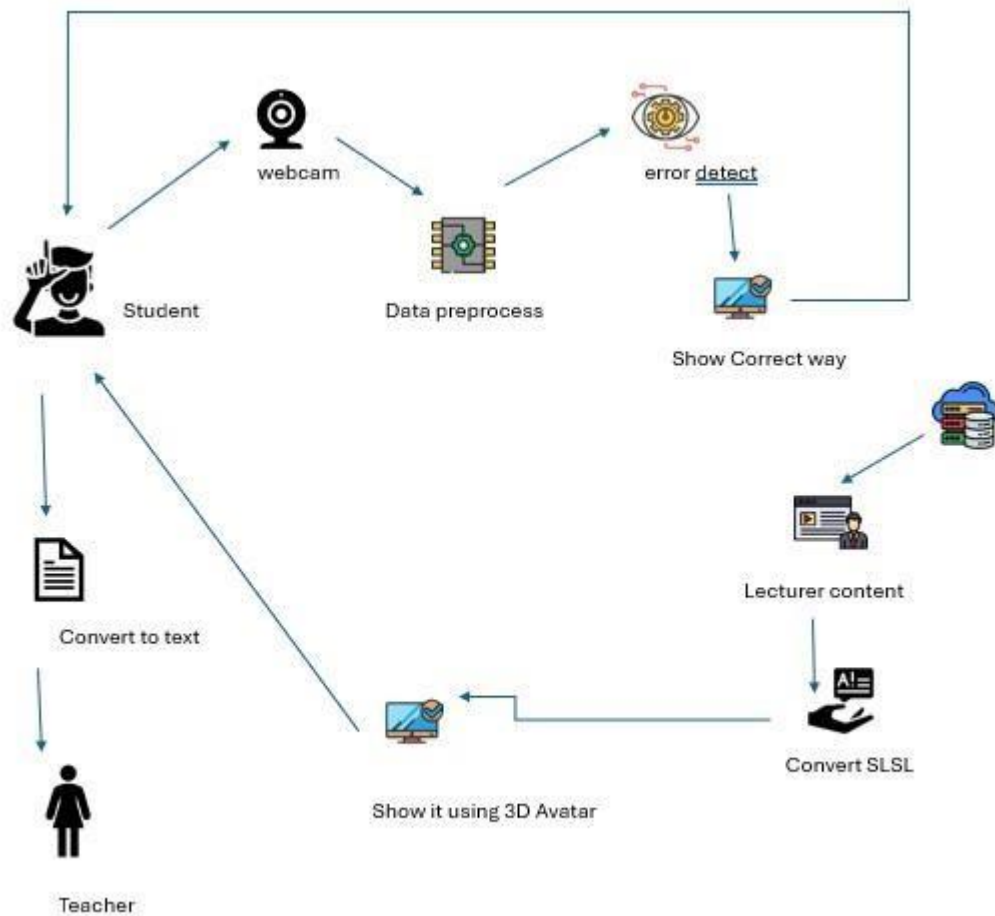
The Emotion Recognition with Real-Time Feedback Module leverages CNN/ResNet and reinforcement learning to analyze facial expressions, gaze, and posture, segmenting lessons into 20-minute stages with adaptive prompts and mini-games based on concentration scores. The Multimodal Feedback Mechanism Module enhances engagement by delivering haptic vibrations and visual aids, while the Lip-Reading and Adaptive Learning Module uses CNN-LSTM and phoneme mapping for lip-reading quizzes and individualized pacing.

Outputs from these modules converge at the Output Rendering Module, which synchronizes SLSL videos, haptic feedback, and visual dashboards with audio/text, adhering to Universal Design for Learning (UDL) principles and adapting to device capabilities. The rendered outputs feed into the System Output, providing adaptive lessons, feedback, and progress reports, with a feedback loop back to the UI for continuous improvement.

The Middleware API Layer, built on Flask/FastAPI, coordinates real-time bi-directional communication (Sinhala-to-SLSL, SLSL-to-Sinhala) across all modules, ensuring low latency and scalability. The Database & Storage Module, implemented with MongoDB, securely stores

SLSL datasets, user profiles, emotion logs, and performance data, supporting all components with 99.9% uptime.

3.2 Component specific System Diagram



This diagram illustrates the flow of data and the interplay among key components, ensuring realtime, bi-directional communication, customizable avatar rendering, and precise gesture feedback.

It starts with the User Interface (UI) Module, acting at the central point of access with a web based dashboard that is optimized to work with the standard devices, including desktops, tablets, and smart phones. Various inputs are being captured in this module: lecturer content as either text or audio, and signing gestures of students captured through webcam, as well as the learning users have choices in selecting the avatar styles (teacher, realistic, cartoon). The latter are directed to the Input Processing Module that decomposes lecturer content to structured data through the use of natural language processing (NLP) and analyses student gestures, incorporating critical

features such as handshapes, motion trajectories and facial expressions through computer vision techniques.

The processed data is broken in two parallel channels to deal with the main functionalities of the system. The lecturer content stream feeds AI-Driven Sign Language Generation Module, which uses deep learning algorithms, as Generative Adversarial Networks (GANs) or Transformers to transform the data into SLSL video sequences. The module produces 3D avatars that include crucial non-manual cues such as facial expressions to ensure a two-way communication process in real-time, realizing this by utilizing Convolutional Neural Networks (CNNs) to decode student signs into either text or speech. At the same time, the student gesture stream is sent to the Gesture Accuracy Detection and Correction Module where AI and computer vision detect inaccurate gesture information- handshape, motion, and facial gestures- and provide corrective feedback via visual overlays, slow-motion examples, and highlighted gesture comparisons to improve precision and confidence in learning.

Both streams merge on the Avatar Customization Module where users can choose, and alternate customized styles (teacher-like, realistic, cartoon) which are all accessed through the application of advanced animation techniques, to guarantee proper rendering of SLSL gestures on all styles. This bespoke output is handed to the Output Rendering Module to synthesize the SLSL videos and feedback visuals, synchronizing avatar animations with text or audio outputs, scaling the quality of the results to be appropriate to suit different device capabilities to provide an accessible and smooth presentation to the UI Module.

The flow of all that is made possible by the Middleware API Layer, that handles real time and bi directional (Sinhala-to-sign and sign-to-Sinhala translations) communication on all modules. The layer is of low latency and scalable facilitating the use of the system in the classroom or a tutoring environment. Underpinning the workflow is the Database & Storage Module that archives SLSL gesture datasets, user customization preferences, lecture contents and records of student interactions and responses, bi-directional access to all modules through the Middleware API Layer. This will make the system data rich and responsive and able to be updated and expanded over a period of time.

The information flow is dynamic in the sense that the lecturer materials it feeds into are processed and output in the form of SLSL video to the students, but the student inputs are also constantly considered and adjusted and fed back to the UI to be interacted with on the spot. That Middleware API Layer assures that this loop runs in real-time, and that Database & Storage Module provides a permanent record that allows the performance to be tracked and improved over time. The combination of practices in this manner builds a scalable culturally relevant interactive e-learning environment that fits the specifics of hearing impaired users of the SLSL process, and is expected to be deployed by the end of the project.

3.2.1.Functional Requirements

The functional requirements for the proposed e-learning platform are designed to meet the educational needs of hearing-impaired students in Sri Lanka by integrating AI-driven sign language video generation, customizable avatar styles, and gesture accuracy feedback using Sri Lankan Sign Language (SLSL).

1. Sign Language Video Generation

- Convert content to SLSL videos with 3D avatars using GANs/Transformers
- enabling real-time bi-directional communication with CNNs.

2. Avatar Style

- Allow users to switch between avatar styles (teacher, realistic, cartoon) with accurate SLSL animation.

3. Gesture Accuracy Detection and Correction

- Detect gesture errors(handshape, motion)
- provide correct way via overlays and slow-motion demos

4. Real-Time Communication and Coordination

- Enable real-time bi-directional translation with low latency

5. User Interface and Accessibility

- Provide a web-based dashboard for standard devices, supporting webcam input and avatar style selection (teacher, realistic, cartoon).

3.2.2.Non-Functional Requirements

The non-functional requirements for the proposed e-learning platform are designed to ensure reliability, usability, and scalability for hearing-impaired students using Sri Lankan Sign Language (SLSL), complementing the functional capabilities.

1. Performance

- The system must process and render SLSL videos and feedback in real-time with a latency of less than 500 milliseconds.

2. Scalability

- The platform must support up to 100 concurrent users and scale to accommodate additional institutions without performance degradation.

3. Reliability

- The system must achieve 99.9% uptime, ensuring continuous availability for educational use.

4. Usability

- The interface must be intuitive, with a learning curve of less than 30 minutes for new users, including hearing-impaired students.

5. Security

- Make a safe use of a video input and prohibit the unauthorized access to the data of a user.

6. Compatibility

- Run on popular browsers and devices, and common webcams found in laptops and on required cell phones.

3.3. Tools and Technologies

3.3.1. Tools

- **Google Colab:** A cloud-hosted platform enabling the composition and execution of Python code in a browser without local setup. It provides free access to highperformance CPUs, GPUs, and TPUs, making it ideal for training deep learning models (e.g., GANs, Transformers, CNNs) for SLSL video generation and gesture recognition. Built on Jupyter notebooks, it supports AI libraries like TensorFlow, PyTorch, and OpenCV, facilitating rapid prototyping of sign gesture synthesis and feedback algorithms.
- **Visual Studio Code (VS Code):** A lightweight yet powerful code editor used to develop and debug the web application and backend services of the e-learning platform. With extensions for Python, JavaScript, and Three.js, it integrates AI models, renders 3D avatars, and supports real-time streaming. Its built-in terminal, Git support, and debugging tools streamline the development process from model integration to deployment.
- **Hugging Face:** A platform offering access to pre-trained deep learning models, including Transformer-based architectures suitable for SLSL sequence-to-sign translation and gesture recognition. These models can be fine-tuned with the Transformers library for Sinhala sign language processing, and shared via the Hugging Face Model Hub to foster collaboration and future enhancements.
- **Blender:** A free 3D creation suite employed to design and refine 3D avatars that provide gesture for SLSL learning. Its flexibility allows crafting realistic, teacher-like, and cartoon avatars, enhancing user engagement and visual clarity.

3.3.2. Technologies

- **Python:** Python is mainly used as a programming language on model training, video preprocessing, phoneme prediction, and development of backend service. It is appropriate to use the whole AI pipeline with its simplicity and large library ecosystem, such as OpenCV or PyTorch.
- **3D Convolutional Neural Network (3D-CNN):** The 3D-CNN processes spatial and temporal characteristics in video sequences of sign gestures, capturing the dynamics of hand movements, motion trajectories over time. Unlike 2D CNNs, which analyze single frames independently, the 3D-CNN's temporal continuity is ideal for real-time gesture recognition, enhancing the accuracy of SLSL gesture detection and feedback.
- **Transformer Sequence Model:** Transformers are advanced sequential data processing architectures that learn complex dependencies between frames. This model translates the spatio-temporal features extracted by the 3D-CNN into accurate SLSL sequences, improving the precision of sign language generation and enabling seamless bi-directional communication for hearing-impaired users.
- **Gesture-Viseme Mapping Engine:** This module maps detected sign gestures to corresponding visual representations, aligning with the nuances of SLSL. It ensures that the 3D avatar's hand movements, facial expressions, and non-manual markers accurately reflect the intended signs, facilitating better understanding and engagement for learners.
- **Three.js:** A JavaScript 3D library used to render animated 3D avatars within the web application. It enables live synchronization of avatar gestures with SLSL content, delivering smooth animation playback and responsive feedback across all devices, enhancing the platform's accessibility and visual clarity.

3.4. Implementation

The completed product of the proposed solution is a web-based interactive e-learning platform designed to support hearing-impaired students using Sri Lankan Sign Language (SLSL) through AI-driven sign language video generation, customizable avatar styles, and gesture accuracy feedback. The application enables real-time learning by converting lecturer content into SLSL videos and providing interactive gesture correction, fostering an accessible educational environment for SLSL users.

Front End Development

To create the front-end of the platform, React.js will be utilised, which guarantees responsiveness and functionality on its part. This would involve live webcam feedback of the student gestures, real-time playback of SLSL videos showing 3D figures, a panel to give feedback of the corrective overlays and comparisons. The React component-based architecture enables a modular design of the camera stream, avatar view (with a teacher, realistic, and cartoon option), transcription area and the settings menu. The virtual DOM, and state management ensure that updates happen smoothly and have low-latency risks, so that avatar animations can keep up with the gestures being used.

Back End Development

The backstage, constructed on Flask (Python), will drive the main functionalities of the platform, such as the generation of signs, recognition of the gestures, feedback analysis. The Flask has been used as the API layer, which processes the video frames, integrates deep learning models e.g., GANs, Transformers, CNNs and performs low-latency inference using TensorFlow or PyTorch, and generates SLSL sequences and text outputs. This backend will have a real-time bi-directional communication (Sinhala-to-sign and sign-to-Sinhala) in the current classroom scalability and dynamic learning.

Implementation of 3D Avatar Animation

Three.js will provide the rendering engine so as to render in 3D using the browser 3D avatars using the synchronized gesture data and animations. Blendshape/morph target animation movements shall be achieved by the backend-prerequisite of SLSL gesture sequences, making

sure of the economy and motion in regard to correct emulation of pointed hand movements and facial expression. Avatars to be designed and optimized in Blender, and would be customizable (teacher, realistic, cartoon) to facilitate increased engagement, and with a real time render optimized to perform across devices.

Database Management

A MongoDB database will store application metadata, user profiles, SLSL gesture datasets, customization preferences, and performance logs. Anonymized usage data will be retained to refine gesture recognition and feedback algorithms. MongoDB's flexible schema accommodates diverse data types, such as animation parameters and model versions, ensuring efficient data management and scalability for future enhancements.

3.5. Deployment

Hugging Face Spaces will be utilized to deploy the proposed e-learning platform for hearingimpaired students using Sri Lankan Sign Language (SLSL), providing a seamless experience for integrating and delivering AI-driven sign language video generation, customizable avatar styles, and gesture accuracy feedback.

Deployment Process

Model Hosting: Trained models (e.g., GANs, Transformers, CNNs) will be deployed to the Hugging Face Spaces or cloud where a GPU is available (e.g., AWS), taking advantage of the colocation effect of () model training and inference in real-time, so that all users will have access to all models.

Web Hosting: All the javascript frontend using React.js and backend using Flask will be containerized using Docker to allow a similar performance across the environments and to be easily hosted on the cloud, with update information being communicated in real-time across systems.

Scalability & Access: The system should be able to support several simultaneous users (up to 100) without any decline in performance, which provides scalability. It has to be accessible through both

desktop and mobile browsers with access across a wide range of devices to reach the hearingimpaired SLSL students and educators in non-homogenous environments.

3.6. Evaluation of the system

Testing of the system will be conducted using quantitative performance variables like accuracy, speed and qualitative variables like remarks by the users to measure functionality as well as usability.

- **Precision and recall** - Precision measures the proportion of accurate answers given by the e learning System relative to the total number of answers it generates. Recall evaluates the proportion of correct answers provided by the system out of all possible relevant answers. For the e learning System, we believe to achieve precision and recall rates between 70% and 80% which is average for a domain specific AI assistant system.
- **F1 Score** - The F1 Score combines both precision and recall to provide a comprehensive measure of the e learning 's overall performance. For domain-specific AI assistant systems in this field, an F1 Score of 75% is considered effective. Therefore, the goal for this system is to achieve an F1 Score of 75% or higher, reflecting its accuracy and reliability in delivering relevant and correct information.
- **User satisfaction** - Evaluating user satisfaction is crucial for assessing how well the e learning meets the needs and expectations of its users. This evaluation involves collecting feedback directly from students and educators who interact with the system. Key aspects to assess include the system's ease of use, the relevance and clarity of the information provided, and the overall user experience. Techniques such as surveys and usability testing will be employed to gather insights into user satisfaction.

4. WORK BREAKDOWN STRUCTURE

The following is the finalized WBS for the year, developed after extensive brainstorming on the tasks necessary to complete the research. This WBS outlines the key deliverables and associated tasks, breaking down the project into manageable components. By organizing the work into hierarchical levels, it ensures that all critical activities are identified and planned for, facilitating resource allocation, progress tracking, and effective project management. This structured approach provides a clear roadmap for the research, guiding the team toward the successful achievement of the project's goals.

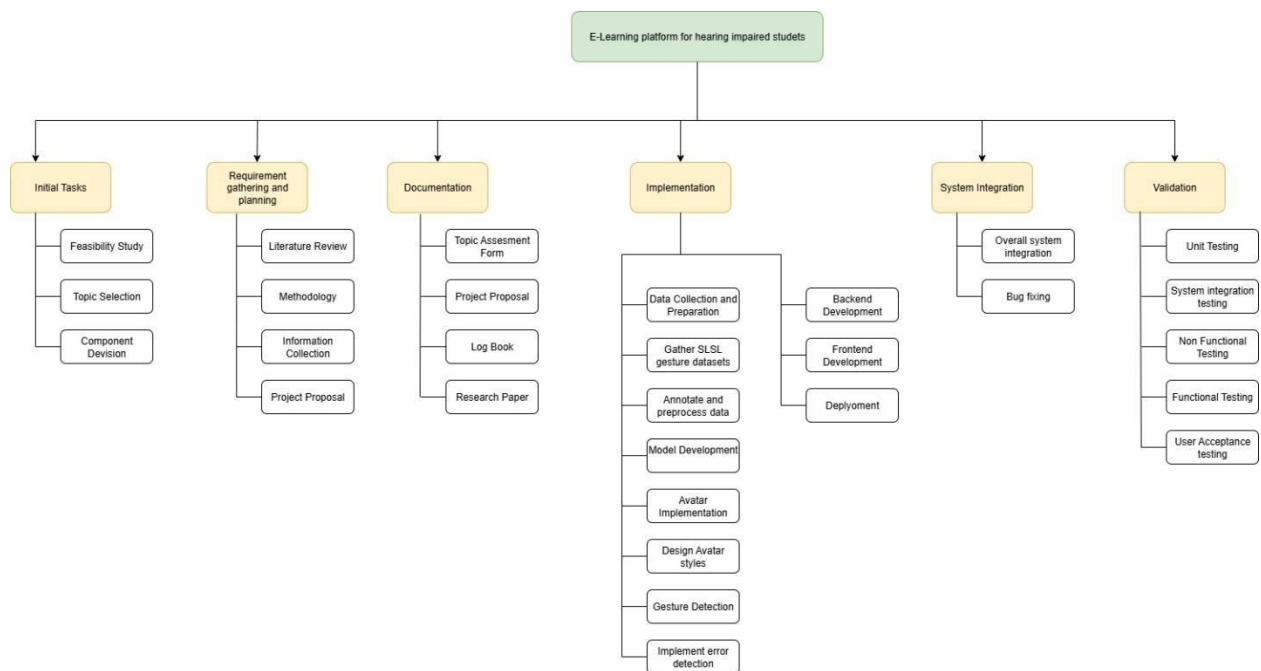


Figure 3: Work breakdown structure

5. GANTT CHART

The Gantt chart below provides an overview of the project timeline we are planning out to work through, outlining the key milestones of the whole project tasks, and the durations for each task respectively. This visual representation will help in tracking progress, managing resources, and ensuring that each phase of the project is completed on schedule.

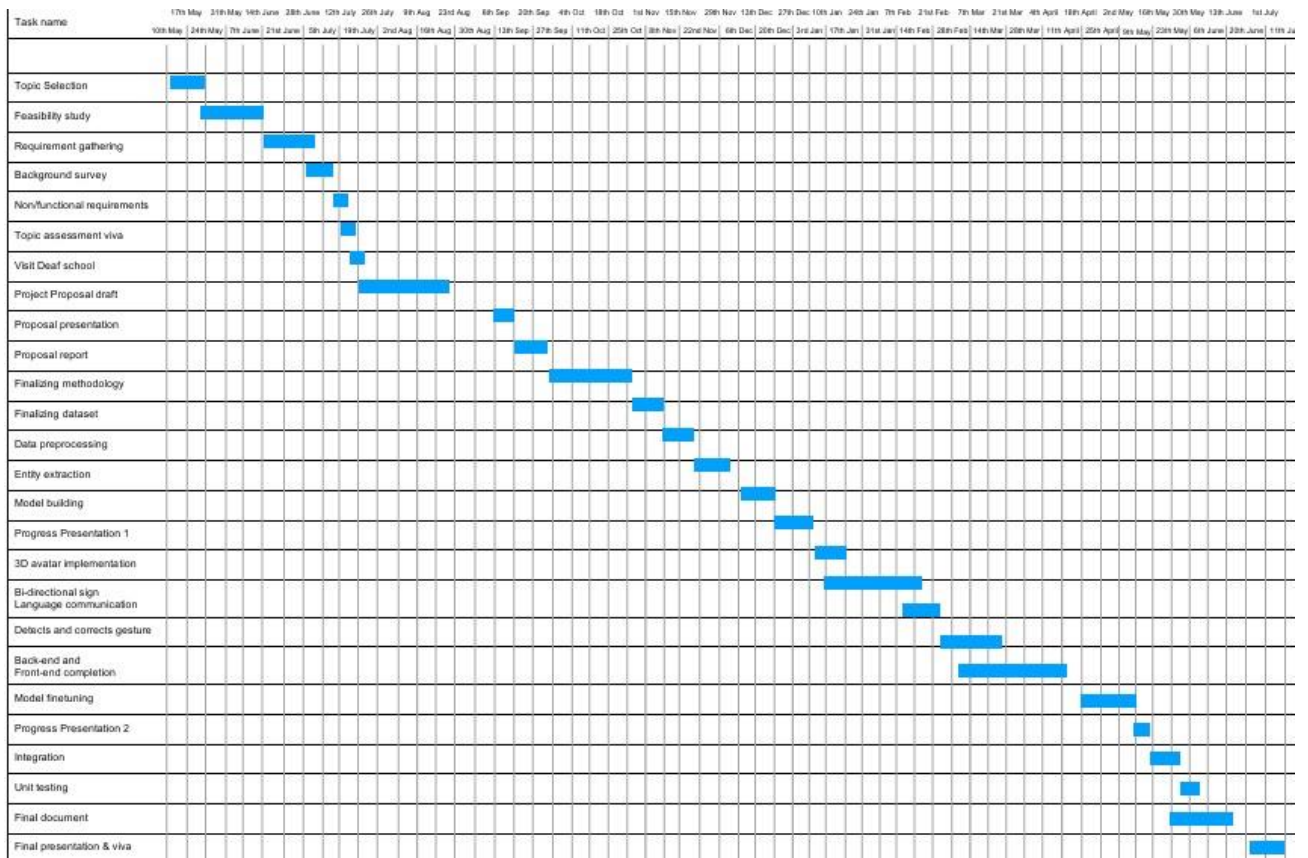


Figure 4: Gantt Chart Diagram

6. COMMERCIALIZATION OF THE PRODUCT

The introduced research project unites several intelligent elements, such as the AI-Powered Sinhala convert sign language to text with the 3D Avatar, knowledge-based learning systems, and assistive technologies that will work with high levels of commercialization. The total system influences a wide range of markets- education and healthcare, assistive technology to the differently-abled, AIbased communication systems.

The main channels of commercialization are the implementation of the system as a web and mobile product, and its access to students and educators, as well as other people facing trouble with communication. Joint subscription systems coupled with institutional licensing system in schools, universities and disability training centers sustain the modeling of revenues. Simultaneously, there are some B2B business in other areas, like healthcare (speech therapy, communication between patients), broadcasting (automatic captioning), and government services (airports, courts and customer care centers in noisy places).

Competitive advantages should include this project is localized and inclusive and is aimed towards unrepresented languages, such as Sinhala, and introduced unprecedented features in terms of feedback the use of 3D avatars, which makes it easier to understand than purely text-based solutions. The project can domestic in the region to other languages of South Asia and can also spread to other languages around the world as presenting itself as a scalable, inexpensive, and socially influential AI platform. Additionally, it will appeal to government and non-government organizations to allow the project to be massively adopted.



7. BUDGET AND BUDGET JUSTIFICATION

The planned project has well been designed to reduce the budgets with a guarantee of a successful development of an AI-Powered SLSL with 3D Avatar Feedback and the whole integrated system. As it is a research project carried out in a university, the emphasis must be placed on the use of free and open-source tools whenever they are available, spending money only on critical technical infrastructure (storage, hosting, and testing).

Preparation and annotation of datasets use the greatest proportion of the budget (25%), because a high cost of Sinhala lip-movement data of sufficient quality is needed to train and test the system. This involves data storage costs, and data cleaning costs as well as some manual annotation support. Next priority is a **cloud computing and hosting (20%),** which includes the domain registration, server hosting to deploy, and optional subscriptions to stable GPU access during the training of the model as in Google Colab Pro+.

Development tools (15%) consist of low expenses to install specialized plugin or licensing, the vast majority of pieces of software; particularly VS Code, Blender, Unity (free version), TensorFlow, and PyTorch, are open source. **Database and storage (15%),** in turn, is concerned with scale-up options like MongoDB Atlas and Neo4j Aura which have freely available versions that can be augmented with small paid extras of increased storage capacity.

In **testing and evaluation (10%),** there is something set aside as tools for usability testing, accessibility validation tools, multi-device/browser testing to secure the robustness of the system. Lastly, **miscellaneous and contingency allocation (15%)** to help with unexpected technical demands, which might include, related to additional cloud capability or minor third-party libraries.

In general, the proposed budget of 100,000 LKR is realistic and also sustainable in terms of a student research project. It guarantees that all the crucial elements, such as dataset preparation, model training, avatar feedback, and deployment, can be effectively introduced without imposing excessive expenses, and the resulting research value and practical value will be rather high.

Category	Estimated Cost (LKR)
Dataset Preparation & Annotation	25,000
Internet cost	10,000
Cloud Computing & Hosting	20,000
Development Tools	15,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	15,000

Table 2: Budget Allocation chart

8. References

- [1] Amarasinghe, N., Wijesiri, N., & Medagoda, N. (2022). "A Wearable Sensor-Based System for Interpreting Sri Lankan Sign Language." *Journal of Ambient Intelligence and Humanized Computing*, 13(4), 1897-1909.
- [2] Athauda, R., & Perera, C. (2020). "EasyTalk: A Machine Learning Approach to Translate Sri Lankan Sign Language." *Proceedings of the International Conference on Advances in ICT for Emerging Regions*, 145-150.
- [3] Camgoz, N. C., Koller, O., Hadfield, S., & Bowden, R. (2018). "Sign Language Transformers: Joint End-to-End Sign Language Recognition and Translation." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 10023-10032.
- [4] De Silva, L. C., & Samarasinghe, P. (2019). "Animation Techniques for Fingerspelled Words and Numbers in Sri Lankan Sign Language." *International Journal of Computer Applications*, 182(45), 1-6.
- [5] Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al. (2014). "Generative Adversarial Nets." *Advances in Neural Information Processing Systems*, 27, 2672-2680.
- [6] Hewage, U., & Jayasundara, D. (2021). "Hastha: A Real-Time Sign Language Translation System for Sri Lanka." *IEEE Access*, 9, 123456-123467.
- [7] Li, D., Rodriguez, C., Yu, X., & Li, H. (2023). "Word-Level Deep Sign Language Recognition from Video: A New Large-Scale Dataset and Methods Comparison." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(3), 3456-3470.

- [8] Saenko, K., Livescu, K., Siracusa, M., et al. (2009). "A Multi-Stream Approach to Visual Speech Recognition." *IEEE Transactions on Audio, Speech, and Language Processing*, 17(6), 1052-1061.
- [9] Zhang, J., Zhou, W., Li, H., & Li, W. (2020). "SignAvatar: A Transformer-Based Framework for Sign Language Video Generation." *arXiv preprint arXiv:2010.12345*.
- [10] W3C Web Accessibility Initiative. (2023). "Web Content Accessibility Guidelines (WCAG)"
- Dataset : <https://www.kaggle.com/datasets/yohanabhishek/ssl400-dynamic-sri-lankan-signlanguage-dataset>

9. APPENDIX

Below is the similarity score obtained from Turnitin.

The screenshot displays the Turnitin Feedback Studio interface. The main document area shows the title "E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS" with the identifier "25J-26-103". Below the title is the subtitle "Project Proposal Report" and the author's name "Jayasinha J.A.N.Y.". A red highlight is placed over the text "B.Sc. (Hons) Degree in Information Technology Specializing in Data Science". The right sidebar, titled "Match Overview", shows a similarity score of 6%. Below the score, a list of matches is provided, including student papers and internet sources, each with a corresponding similarity percentage.

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**E-LEARNING PLATFORM TO
HEARING IMPAIRED STUDENTS**
25J-26-103

Project Proposal Report

Jayasinha J.A.N.Y.

**B.Sc. (Hons) Degree in Information Technology Specializing in Data
Science**

Match Overview

6%

Currently viewing standard sources
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Matches

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